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ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech Aerospace Engineering
Name of the Paper: Aircraft Stability and Control

Semester: VII
Paper Code: TAS 504

Time: 3 Hours

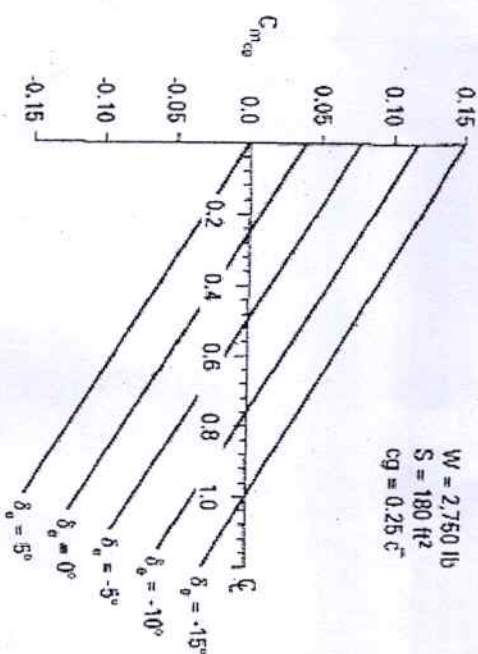
Maximum Marks: 100

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question.
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	(20 marks)
(a) Derive the simplified expression for the pitching moment coefficient of the wing about the center of gravity, starting from the general moment equation and stating the assumptions for small angles of attack.	
(b) The wing-fuselage pitching-moment characteristics of a high-wing, single-engine, general-aviation airplane is given about the center of gravity by $C_{m_{cgw}} = -0.05 - 0.0035\alpha$, where α is the fuselage-reference-line angle of attack (in degrees) and the subscript wf denotes the combined wing-fuselage contribution. The pertinent geometric and aerodynamic data are: $S_w = 17 \text{ m}^2$, $b_w = 11 \text{ m}$, $AR_w = 7.3$, $\frac{c_w}{c} = 1.5$, $i_w = 2.0^\circ$, $C_{l_{\alpha w}} = 0.07$ $= 0.26$, $C_{l_{\alpha wf}} = \frac{0.07}{\text{deg}}$ Design the horizontal tail (planform area and incidence) so that the complete airplane (wing-fuselage-tail) exhibits the following pitching-moment characteristics about the center of gravity: $C_{m_{cgwf}} = 0.15 - 0.025\alpha$ where the subscript wf denotes the combined wing-fuselage-horizontal-tail contribution and α is in degrees. Assume the following data for the horizontal tail: $l_t = 4.6 \text{ m}$, $\eta = 1$, $AR_t = 5$, $C_{l_{\alpha t}} = 0.073/\text{deg}$ Determine: 1. the required horizontal tail area S_t , and 2. the required tail incidence angles i_t .	

(c) An airplane exhibits the following pitching moment characteristics: The slope of the pitching moment coefficient curve with respect to the lift coefficient is $\frac{dC_{m'}}{dC_l} = -0.15$. The pitching moment coefficient at zero lift is $C_{m_0} = 0.08$. The center of gravity of the airplane is located at $\bar{x}_{cg} = 0.30$, where c represents the mean aerodynamic chord of the wing. Based on above data Answer the following question: (i) Explain the concept of static longitudinal stability of an aircraft. What condition must be satisfied for an airplane to be statically stable in pitch? (ii) Using the given data, determine the trim lift coefficient ($C_{l_{trim}}$) at which the pitching moment about the center of gravity becomes zero, corresponding to the trim condition. (iii) Define the term stick-fixed neutral point and explain its physical significance in aircraft longitudinal stability. (iv) Determine the location of the stick-fixed neutral point ($\bar{x}_{np}$) for the given airplane data.	
Q2	(20 marks)
(a) For the data shown below in figure, determine the following I. The stick fixed neutral point. II. If we want to fly aircraft at velocity of 125 ft/sec at sea level, determine the trim lift coefficient and the elevator deflection angle required for trim. Take density $= 0.00237 \text{ slug/ft}^3$.	

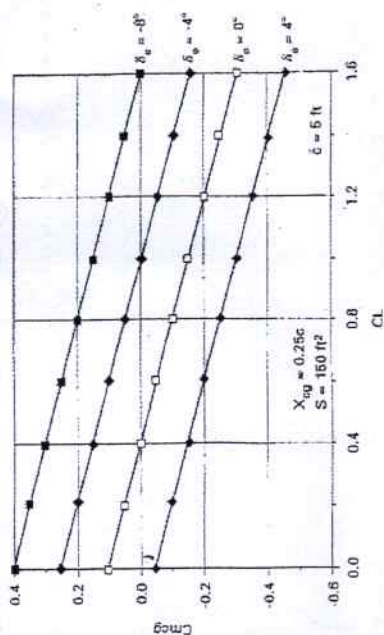


The pitching moment Characteristics of a General Aviation aircraft, with the landing gear and flaps in their retracted positions, are presented in following figure.

I. Determine the location of the Stick fixed neutral point of the aircraft.

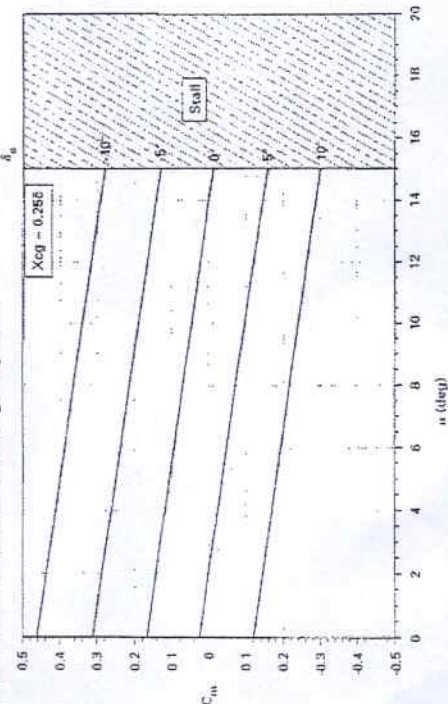
II. If the aircraft weight is 3000 lb and is flying at velocity at 160 ft/sec at sea level, where the air density is $0.002378 \text{ slug/ft}^3$. Determine the elevator deflection angle required for trim.

III. Discuss the variation of pitching moment curve when the landing gear is deployed. Also, explain how the deflection of flaps influences the static stability of aircraft.



The pitching moment coefficient vs angle of attack curve of large transport aircraft is present in following figure. Use the figure and given information to answer the following question.

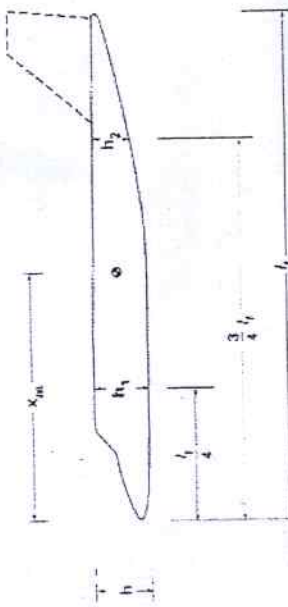
- I. Estimate the stick fixed neutral point of the aircraft.
 $C_L = 0.03 + 0.08\alpha(\text{deg})$
 $-0.15^\circ \leq \delta_e \leq 20 \text{ degree}$
- II. Estimate the elevator control power ($C_{m\delta_e}$)
- III. Determine the forward Center of gravity limit.



Q3 (20 marks)

Design and size the vertical tail of aircraft configuration shown in following figure so that its directional or weathercock stability has a value of $C_{np} = +0.1 \text{ per rad}$. Clearly state all the assumption made during the analysis. Assume the flight velocity of 150 m/sec at sea level.

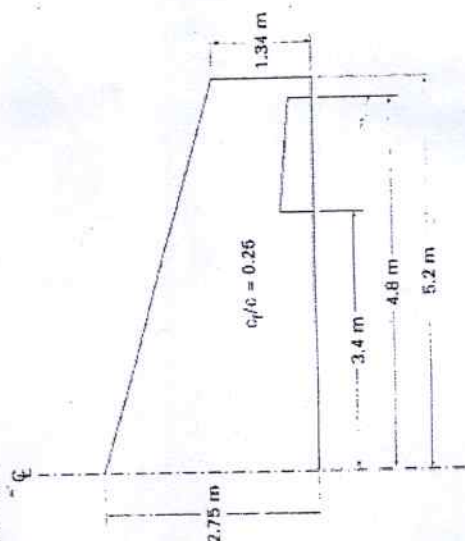
$S = 22 \text{ m}^2$, $b = 10.5 \text{ m}$, $z_w = 0.4 \text{ m}$, $d = 1.6 \text{ m}$, $l_t = 13.9 \text{ m}$, $x_m = 8 \text{ m}$, $w_f = 1.7 \text{ m}$, $S_f = 15.4 \text{ m}^2$, $h = 1.6 \text{ m}$, $h_1 = 1.6 \text{ m}$, $h_2 = 1.07 \text{ m}$



Following figure presents a sketch of a wing planform for a General aviation aircraft.

I. Apply strip theory to determine the roll control power of given aircraft.

II. Discuss the accuracy and limitations of the strip theory integration technique used in question I.

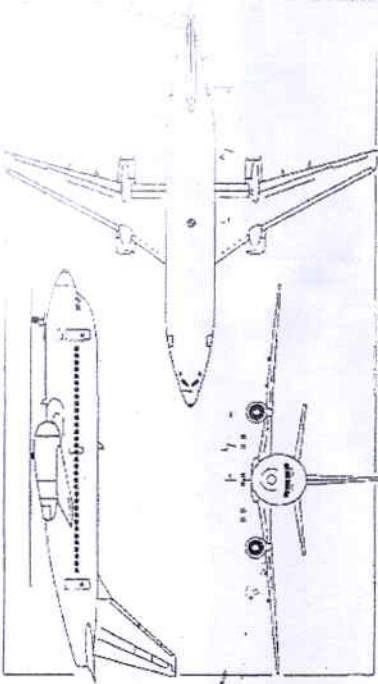


Half Span of Wing

3(C)

For the twin engine aircraft shown in figure, determined the required rudder size to maintain directional control of the aircraft if there is sudden failure of one engine. The flight and geometrical data of the aircraft is provided below.

Wing area (S)	92 m ²
Wing span (b)	25.5 m
Vertical tail area (S _v)	31 m ²
Aspect ratio of vertical tail (AR _v)	4.4
Tail arm (l _v)	11.3 m
Tail efficiency factor (η _v)	1.0
Rudder deflection (δ _r)	+15° to -15°
Thrust per engine	63000 N
Engine lateral distance (Y _T)	4.9 m
Flight velocity (V)	78 m/sec
Air density	1.225 kg/m ³



(20 marks)

Explain in detail the concept of the Body Axis System and Earth-Fixed Axis System used in aircraft motion analysis. Discuss their orientation, significance, and interrelation with the help of neat and clearly labeled sketch.

Explain the effects of gravitational and thrust forces acting on an aircraft with respect to the body-axis coordinate system. Describe how these forces act on the airplane and write the expressions for their components. Support your answer with neat and labeled sketches

Explain the concept of Small-Disturbance Theory as applied to aircraft motion. Discuss the basic assumptions made in developing this theory and describe how the equations of motion are linearized for small perturbations about a steady flight condition.

CO2,
CO3

Q5

(20 marks)

Briefly explain the following aircraft dynamic stability modes with suitable sketches wherever necessary:

- Short Period Mode
- Phugoid Mode
- Dutch roll
- Wing Rock

CO3,
CO4

(b) The root of the characteristics equation for the longitudinal dynamics of a Cargo aircraft are given as:

$$\lambda_1 = -0.02 + 0.2i, \lambda_2 = -0.02 - 0.2i, \lambda_3 = -2.5 + 2.6i, \lambda_4 = -2.5 - 2.6i, \text{ where } i = \sqrt{-1}$$

- Identify the pair of eigenvalues that represent the Phugoid mode and justify your answer
- Identify the pair of eigenvalues that represent the Short Period mode and justify your answer
- For the short period mode, determine natural frequency, damping ratio and damped frequency.

(c) During the analysis of an aircraft's longitudinal dynamics, engineers often study the pitching motion about the center of gravity to understand the airplane's stability and response characteristics. Suppose, during flight testing, it was observed that the airplane exhibits small oscillations in pitch when disturbed from its trimmed condition. To mathematically model this behavior, the pitching motion about the center of gravity is represented by the following second-order differential equation:

$$\ddot{\alpha} + 4\dot{\alpha} + 36\alpha = 0$$

This equation reflects the airplane's dynamic response, including both the stiffness (restoring moment) and damping (aerodynamic resistance) effects. Based on this mathematical model, determine the following parameters for the pitching motion:

- The natural frequency (ω_n) in rad per sec.
- The damping ratio (ξ).
- The damped natural frequency (ω_d) in rad per sec.

